

## RESEARCH INTERESTS

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Machine Learning, Deep Learning, Multilingual and Low-Resource NLP, AI Safety and Guardian Models, Clinical Decision Support Systems, Evaluation of LLMs in Healthcare, AI for Health Systems in LMICs.

## EDUCATION

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| <b>Brown University</b><br>Ph.D. in Computational Biology, Advisor: Hamish Fraser<br>M.S. in Computer Science, GPA: 3.9/4.0 | Providence, USA<br>2023–Present<br>2021–2023 |
| <b>Nelson Mandela -AIST</b><br>M.S. in Information & Communication Science & Engineering, GPA: 4.34/5.00                    | Arusha, Tanzania<br>2012–2014                |
| <b>Saint Augustine University of Tanzania</b><br>B.S. in Computer Science, GPA: 4.50/5.00                                   | Iringa, Tanzania<br>2006–2009                |

## EXPERIENCE

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| <b>Graduate Research Assistant</b><br>Brown University                                                                                                                                                                                                                                                                                                              | Providence, USA<br>2023–Present             |
| <ul style="list-style-type: none"><li>Conduct research on the evaluation, robustness, and safety of machine learning and large language model (LLM)-based clinical decision support systems.</li><li>Develop multilingual and low-resource evaluation benchmarks to assess reliability, hallucination risk, and policy compliance in healthcare settings.</li></ul> |                                             |
| <b>Data Scientist</b><br>Center for Financial Inclusion (Cenfri)                                                                                                                                                                                                                                                                                                    | Remote, South Africa<br>March–November 2018 |
| <ul style="list-style-type: none"><li>Led the development of a machine learning-based credit scoring system to improve financial access for underserved rural communities in Tanzania.</li></ul>                                                                                                                                                                    |                                             |
| <b>Assistant Lecturer</b><br>College of Business Education                                                                                                                                                                                                                                                                                                          | Dodoma, Tanzania<br>2015–2021               |
| <ul style="list-style-type: none"><li>Taught undergraduate courses in programming (C/C++), business mathematics, and supervised capstone research projects.</li><li>Mentored students in applied machine learning and software development projects.</li></ul>                                                                                                      |                                             |
| <b>Teaching Assistant</b><br>Ruaha Catholic University                                                                                                                                                                                                                                                                                                              | Iringa, Tanzania<br>2009–2010               |
| <ul style="list-style-type: none"><li>Supported instruction in computer science courses and assisted with laboratory sessions and grading.</li></ul>                                                                                                                                                                                                                |                                             |

## PUBLICATIONS

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- [1] T. Abdullahi, **M. Mgonzo**, M. Oduwale, P. Okewunmi, A. Owodunni, R. Singh, and C. Eickhoff, “Ubuntuguard: A culturally-grounded policy benchmark for equitable ai safety in african languages”, *AAAI Workshop on Language Models for Underserved Communities*, 2026.

- [2] **M. Mgonzo**, K. Oketch, N. A. Etori, W. Mang’eni, E. Nyaki, and M. S. Mollel, “Learning from scarcity: Building and benchmarking speech technology for sukuma”, *EACL Workshop on Language Models for Low-Resource Languages*, 2026.
- [3] R. Esfandiarpour, G. Zerveas, R. Zhang, **M. Mgonzo**, C. Eickhoff, and S. Bach, “Beyond contrastive learning: Synthetic data enables list-wise training with multiple levels of relevance”, in *EMNLP*, 2025.

## PROFESSIONAL SERVICE

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- **Reviewing**

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|---------------------------------------------------|------------|
| Machine Learning for Healthcare (ML4H) Conference | 2023–2025  |
| Black in AI (BAI) Workshop                        | 2024       |
| Deep Learning Indaba (DLI) Conference             | 2024, 2025 |
- **Volunteering**

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| SIGIR | 2022 |
| FAccT | 2022 |
- **Organizing**

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| IndabaX, Nelson Mandela AIST, Arusha, Tanzania | 2020 |
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## TEACHING

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- **Graduate Teaching Assistant** at Brown University Fall 2024  
*Evaluation of Health Information Systems (BIOL 2075)*
- **Assistant Lecturer** at College of Business Education 2010–2021  
*Advanced Business Mathematics, Programming in C, C++ (SMU 7304, TT5102 & TT6102)*

## SKILLS

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- **Programming:** Python (advanced), C/C++, SQL.
- **ML & AI:** PyTorch, TensorFlow, HuggingFace, Scikit-learn.
- **Developer Tools:** Git, Linux, LaTeX.
- **Big Data Analytics Frameworks:** Hadoop, MapReduce.

## LANGUAGES

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- **Language: English** Fluent
- **Language: Swahili** Native Speaker

## SCHOLARSHIPS AND AWARDS

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- Black in AI Travel Award (NeurIPS 2022) 2022
- Google CS Research Mentorship Program (CS-RMP) 2021
- ACM Travel Grant (FAccT 2019, FAccT 2020) 2019, 2020
- Google Conference Travel Award(DLI 2018) 2018